# Robust Balancing and Trajectory Control of a Self-Driving Bicycle

T.-J. Yeh, *Member, IEEE*, Tzu-Chieh Lin, and Alexander Chia-Bin Chen

***Abstract*— The purpose of this research is to construct a self-driving bicycle that can balance itself and automatically track a designated trajectory in campus environments. For balancing, a lower level controller is designed based on the dynamic model of the bicycle. It allows the bicycle to achieve lateral stability and cornering action with robustness to speed variations. The design methodology adopts a linear-parameter-varying (LPV) approach by first decomposing the dynamic model into a convex combination of four linear subsystems with time-varying coefficients and then solving a set of linear matrix inequalities (LMIs) to compute the gain matrix for robust state feedback. For trajectory tracking, a high-level controller is designed using similar LPV approach. It allows the bicycle to robustly follow a pregenerated virtual vehicle motion on a given path regardless of the speed and yaw-rate changes of the virtual vehicle along the path. The control system is verified both numerically and experimentally on a prototype bicycle. In particular, the experiment shows that the self-driving bicycle can follow the testing route in campus with rms error less than 18 cm.**

***Index Terms*— Balancing control, bicycle, linear parameter varying (LPV), sensor fusion, trajectory control.**

## I. Introduction

AS THE least expensive means of wheeled transportation, bicycles are widely used for many activities, such as commute, sport, recreation, and so on. Bicycles are considered to be environmentally friendly, because they can reduce traffic congestion and air pollution in urban areas. The recent introduction of electric bicycles can further enhance the range and mobility of bicycles. From system dynamics perspective, bicycles are in the category of wheeled-inverted-pendulum vehicles and exhibit interesting dynamic behavior. Modeling, analysis, and control of bicycles, thus, have attracted significant attention in research community ever since they were invented.

Regarding the control studies for bicycles, the recent advances in digital computers, sensor, and actuator technologies have drawn significant research interests on developing self-balancing bicycles. For instance, Cerone et al. [1] exploited linear-parameter-varying (LPV) nature of the bicycle model proposed in [2] to design a control system that automatically balances a riderless bicycle using the steering torque. Their control problem is formulated as the design of an LPV state-feedback controller that guarantees stability when the speed varies within a given range and its derivative is bounded. Huang et al. [3] developed a miniaturized humanoid robot to ride and pedal a bicycle of comparable size. The robot balances and steers the bicycle via controlling the angle of the handlebar. The controller, which is designed based on a constant-speed bicycle model, can automatically counteract the mass imbalance in the robot-bicycle system and allow it to perform straight-line steering and cornering. In [4], two nonlinear controllers are designed to balance a bicycle, respectively, by the flywheel and by the handlebar. Global stability for both cases is theoretically proved using Lyapunov theory. Wang et al. [5] therein further combined the two controllers using the gain scheduling approach to accomplish balancing for both stationary and dynamic cases.

There are also researchers developing methods for trajectory control with simultaneous balancing for bicycles to achieve autonomy. The methods are based on decomposing the bicycle dynamics into an external–internal convertible (EIC) structure in which the external subsystem performs trajectory tracking and the internal subsystem performs balancing [6]. Various nonlinear control techniques, including the Lyapunov design [7], sliding-mode control [8], and feedback linearization [9], have been employed to design controllers for, respectively, the external and the internal systems. In order for the control system to comply with the constraints on the roll, steer, and heading angle, bicycle's position, and so on, there are studies applying model predictive control (MPC) to stabilize the internal subsystem [10] and to perform trajectory tracking for the external subsystem [11], [12]. The EIC-based approaches basically ignore the dynamic coupling between internal and external subsystems and design the tracking and balancing controllers separately, so only approximate trajectory-tracking problem is solved. To tackle the coupling effect, Cui et al. [7] applied the small-gain technique to tune the gains for the tracking and balancing controllers, so that overall stability of the closed-loop bicycle system can be theoretically guaranteed.

This brief is devoted to developing practically viable control methods for self-driving bicycles. The theoretical contribution of the developed methods is that by formulating both balancing control and trajectory-tracking control as stabilization/regulation problems for LPV systems, a control framework is proposed for synthesizing both controllers with robustness: while the balancing control is robust against the bicycle's speed variations, the tracking control is robust against the commanded speed and yaw-rate variations along the reference path. Compared with the robust LPV approach proposed in [1] in which solving an infinite family of linear matrix inequalities (LMIs) is required and the resultant controller only balances the bicycle in the upright position, the approach proposed here solves only four LMIs and can also be applied

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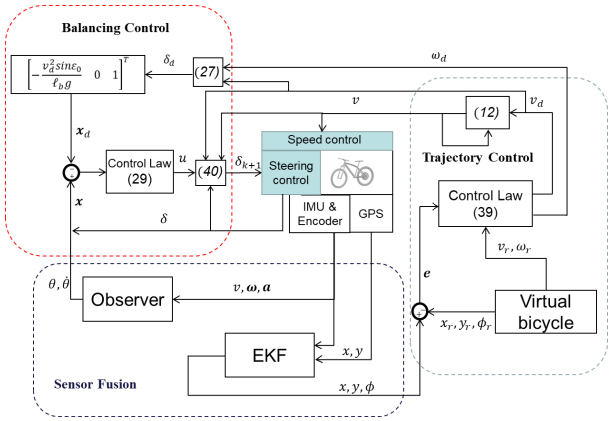



Fig. 1. Control block diagram of the self-driving bicycle.

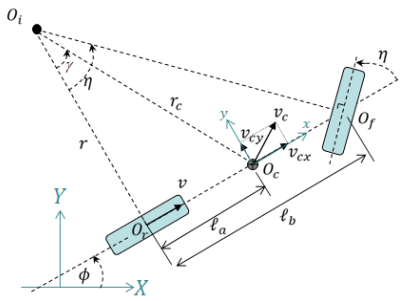


Fig. 2. Top view of a bicycle.

to the cornering case. As for the practical contribution, it is noted that the existing studies on bicycle control mostly rely on simulations or miniaturized prototypes to validate the theories [3], [4]. Among few studies, which provide experimental demonstrations using real-size bicycles, either only balancing control was considered [1], [13], or tracking performance under simultaneous balancing was examined using small-scale trajectories of selected geometric shapes [9]. Because of the systematic design procedure and the control simplicity, the developed methods can be applied experimentally on a real-size bicycle to perform large-scale self-driving tests on campus roads. In particular, by incorporating a sensor fusion algorithm for localization purposes, the self-driving bicycle is able to overcome real-world problems, including uneven pavement, GPS obstruction, and so on. The block diagram showing how the proposed balancing control, trajectory control, and sensor fusion algorithms are integrated with the bicycle's sensing and actuation is depicted in Fig. 1.

This brief is organized as follows. An LPV model that describes the speed-dependent lateral dynamics of the bicycle is given in Section II. Section III proposes a control design methodology, which can provide robust stability to LPV systems. In Section IV, the kinematic model for trajectory-tracking control is also converted into the LPV form, and the proposed control methodology is applied to design robust tracking controller. A sensor fusion algorithm for localization is presented in Section V. The self-driving control performance is validated both numerically and experimentally in Section VI. Finally, conclusions are given in Section VII.

## II. Modeling of the Lateral Dynamics

The modeling for the lateral dynamics of a bicycle is adopted from [3]. To begin with, the kinematic relations among crucial motion variables are analyzed.

### A. Kinematic Analysis

Fig. 2 shows the top view of a bicycle in which $O_c$ is its center of gravity (COG), $O_f$ is the center of the front wheel, and $O_r$ is the center of the rear wheel. Assume both wheels have no lateral slip, the rear wheel is driven with speed $v$, and the front wheel is steered with a directional angle $\eta$. Driving and steering actions make the bicycle turn with a yaw angle $\phi$ with respect to the inertial frame $XY$. Under the no-slip condition, the instantaneous center of rotation, denoted by $O_i$, is the intersection of the two extension lines, respectively, from the axles of the front and rear wheels. Letting $\ell_a = \overline{O_r O_c}$, $\ell_b = \overline{O_r O_f}$, $r = \overline{O_i O_r}$, $r_c = \overline{O_i O_c}$, and $\gamma = \angle O_r O_i O_c$, the following trigonometric relations hold:

$$r_c = \frac{\ell_a}{\sin\gamma} \tag{1}$$

and

$$r = \frac{\ell_a}{\tan\gamma} = \frac{\ell_b}{\tan\eta}. \tag{2}$$

The yaw rate of the bicycle, $\dot{\phi}$, is equal to $(v/r)$. From the second equality in (2), $\dot{\phi}$ can be written as follows:

$$\dot{\phi} = \frac{\tan\eta}{\ell_b} v. \tag{3}$$

Let $v_c$ be the magnitude of the bicycle's velocity at the COG $O_c$, and $v_{cx}$ and $v_{cy}$ be the components of $v_c$, respectively, along the $x$- and $y$-axes of the body frame defined in Fig. 2. Since the bicycle body is rigid, we have $v_{cx} = v$, and $v_c$ and $v_{cy}$ are related to $v$, respectively, by

$$v_c = r_c\dot{\phi} = \frac{\ell_a \tan\eta}{\ell_b \sin\gamma} v \tag{4}$$

and

$$v_{cy} = v_c \sin\gamma = \frac{\ell_a \tan\eta}{\ell_b} v. \tag{5}$$

### B. Dynamic Analysis

In the rear view shown in Fig. 3(a), the bicycle is modeled as an inverted pendulum whose mass $M$ is concentrated at the COG. The COG is located at a distance of $h$ from the ground contact point, and the bicycle is tilted with a roll angle $\theta$. When deriving the lateral dynamics, $\theta$ is treated as the generalized coordinate, so the corresponding generalized force is the total moment about the $x$-axis. As shown in Fig. 3(a), forces that contribute to the moment about the $x$-axis include gravity and the $y$-axis components of the inertial force and the centrifugal force, which are, respectively, given by $-M(dv_{cy}/dt)$ and $-M(v_c^2/r_c)\cos\gamma$. By Lagrange

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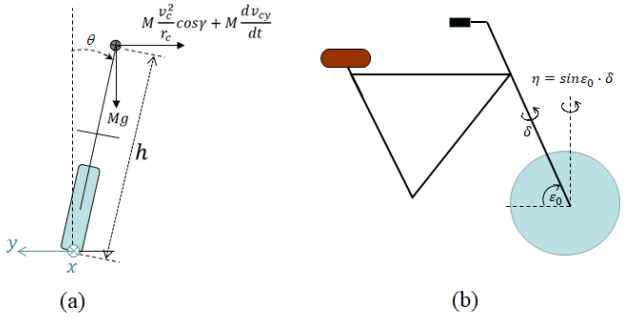



Fig. 3. (a) Rear view and (b) lateral view of a bicycle.

formulation, the lateral dynamics, thus, can be derived as follows:

$$\begin{aligned} Mh^2\ddot{\theta} &= Mh\cos\theta\left(\frac{v_c^2}{r_c}\cos\gamma + \frac{dv_{cy}}{dt}\right) + Mgh\sin\theta \\ &= Mh\cos\theta\left(\frac{v^2}{\ell_b}\tan\eta + \frac{\ell_a v\sec^2\eta}{\ell_b}\dot{\eta}\right. \\ &\quad \left. + \frac{\ell_a\tan\eta}{\ell_b}\dot{v}\right) + Mgh\sin\theta \end{aligned} \tag{6}$$

in which the second equality is due to the substitutions of the expressions of $v_c$ and $v_{cy}$ in (4) and (5)[1] and the relation $\tan\gamma = (\ell_a/\ell_b)\tan\eta$ inferred from (2).

When $\eta$ and $\theta$ are small, (6) can be further linearized as follows:

$$Mh^2\ddot{\theta} \approx Mh\left[\left(\frac{v^2}{\ell_b} + \frac{\ell_a\dot{v}}{\ell_b}\right)\eta + \frac{\ell_a v}{\ell_b}\dot{\eta}\right] + Mgh\theta. \tag{7}$$

According to Fig. 3(b), where the lateral view of the bicycle is partially shown, the directional change of the front wheel is due to the vertical projection of the rotation of the steering handlebar via the caster angle of the front fork assembly. Defining the steering angle of the handlebar by $\delta$ and the caster angle by $\varepsilon_0$, the directional angle $\eta$ can be expressed as follows:

$$\eta = \sin\varepsilon_0 \cdot \delta. \tag{8}$$

Substituting the above expression into (7) leads to

$$\ddot{\theta} - \frac{g}{h}\theta = \frac{\ell_a v\sin\varepsilon_0}{\ell_b h}\left[\dot{\delta} + \left(\frac{v}{\ell_a} + \frac{\dot{v}}{v}\right)\delta\right]. \tag{9}$$

### C. State-Space Model for Control Design

In (9), if both the bicycle speed and the steering angle are set as constants, the resulted open-loop lateral dynamics are unstable due to the inverted pendulum mode at $(g/h)^{1/2}$. As a result, feedback control is needed to modulate the steering angle to stabilize/balance the bicycle under the possible speed variations. For control design purposes, we first convert (9) into a state-space form by defining the state vector $\mathbf{x}$ as $\begin{bmatrix}\theta & \dot{\theta} & \delta\end{bmatrix}^{\mathrm{T}}$ and a new control input as

$$u = \dot{\delta} + \left(\frac{v}{\ell_a} + \frac{\dot{v}}{v}\right)\delta \tag{10}$$

and the state equation of the linearized lateral dynamics of the bicycle is, thus, given by

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \tag{11}$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ \frac{g}{h} & 0 & 0 \\ 0 & 0 & -\frac{v}{\ell_a} - \frac{\dot{v}}{v} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{\ell_a v\sin\varepsilon_0}{\ell_b h} \\ 1 \end{bmatrix}.$$

The major challenge in the feedback control design is that the presence of $v$ and $\dot{v}$ in the $\mathbf{A}$ and $\mathbf{B}$ matrices makes the system dynamics time-varying when the bicycle experiences speed/acceleration changes. To facilitate control design for the time-varying dynamics, we first impose the following rule for the acceleration of the bicycle:

$$\dot{v} = cv(v_d - v) \tag{12}$$

in which $c$ is a positive constant and $v_d$ is the command velocity. Imposing such an acceleration rule allows $v$ to track a slowly varying $v_d$.[2] Moreover, with $(\dot{v}/v) = c(v_d - v)$, all the time-varying terms in $\mathbf{A}$ and $\mathbf{B}$ are linear with respect to $v$ or $v_d$. Assume both $v$ and $v_d$ are bounded below by $v_{\min}$ and bounded above by $v_{\max}$. We define two time-varying parameters $\alpha = (v - v_{\min})/(v_{\max} - v_{\min})$ and $\beta = (v_d - v_{\min})/(v_{\max} - v_{\min})$ to represent the normalized values for $v$ and $v_d$. Expressing $v$ and $v_d$ in terms of $\alpha$ and $\beta$, the state equation in (11) can be transformed into an LPV form as follows:

$$\dot{\mathbf{x}} = \mathbf{A}_0\mathbf{x} + \mathbf{B}_0 u + \alpha\mathbf{A}_\alpha\mathbf{x} + \alpha\mathbf{B}_\alpha u + \beta\mathbf{A}_\beta\mathbf{x} \tag{13}$$

where

$$\mathbf{A}_0 = \begin{bmatrix} 0 & 1 & 0 \\ \frac{g}{h} & 0 & 0 \\ 0 & 0 & -\frac{v_{\min}}{\ell_a} \end{bmatrix}, \quad \mathbf{B}_0 = \begin{bmatrix} 0 \\ \frac{\ell_a\sin\varepsilon_0 v_{\min}}{\ell_b h} \\ 1 \end{bmatrix}$$

$$\mathbf{A}_\alpha = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\left(\frac{1}{\ell_a} - c\right)(v_{\max} - v_{\min}) \end{bmatrix}$$

$$\mathbf{B}_\alpha = \begin{bmatrix} 0 \\ \frac{\ell_a\sin\varepsilon_0(v_{\max} - v_{\min})}{\ell_b h} \\ 0 \end{bmatrix}$$

and

$$\mathbf{A}_\beta = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -c(v_{\max} - v_{\min}) \end{bmatrix}.$$

## III. Robust Balancing Control

With the lateral dynamics in the LPV form, it is desired to use the physical states $\theta$, $\dot{\theta}$, and $\delta$ to implement full-state-feedback control to robustly stabilize and regulate the system under the bounded and time-varying parameters $\alpha$ and $\beta$.

[1]The kinematic analysis implicitly assumes that the bicycle is in the upright position. The tilt of the bicycle will introduce errors to the derived equations. The errors are first-order effects and can be neglected if the tilt is small.

[2]Given a constant $v_d$ and an initial velocity $v(0)$, the solution to (12) is $v(t) = (v_d/1 + ((v_d/v(0)) - 1)e^{-cv_d t})$, so $v(t)$ monotonically converges to $v_d$, as $t \to \infty$ and $c$ can be used to tune the rate of convergence.

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To do so, we follow the control approach proposed in [14] to transform the LPV system into a "polytopic form" as follows:

$$\dot{\mathbf{x}} = \sum_{k=1}^{4} \rho_k(t)\left(\mathbf{A}_k\mathbf{x} + \mathbf{B}_k u\right) \tag{14}$$

using the following lemma.

*Lemma 1:* The dynamic model in (13) can be transformed into the polytopic form in (14) with

$$\rho_1(t) = (1-\alpha)(1-\beta) \tag{15}$$
$$\rho_2(t) = \alpha(1-\beta) \tag{16}$$
$$\rho_3(t) = (1-\alpha)\beta \tag{17}$$
$$\rho_4(t) = \alpha\beta \tag{18}$$

and

$$\begin{aligned}
&\mathbf{A}_1 = \mathbf{A}_0, \quad \mathbf{B}_1 = \mathbf{B}_0\\
&\mathbf{A}_2 = \mathbf{A}_0 + \mathbf{A}_\alpha, \quad \mathbf{B}_2 = \mathbf{B}_0 + \mathbf{B}_\alpha\\
&\mathbf{A}_3 = \mathbf{A}_0 + \mathbf{A}_\beta, \quad \mathbf{B}_3 = \mathbf{B}_0\\
&\mathbf{A}_4 = \mathbf{A}_0 + \mathbf{A}_\alpha + \mathbf{A}_\beta, \quad \mathbf{B}_4 = \mathbf{B}_0 + \mathbf{B}_\alpha.
\end{aligned}$$

*Proof:* Because the definitions of $\rho_{(.)}$ in (15)–(18) lead to $\rho_2 + \rho_4 = \alpha$, $\rho_3 + \rho_4 = \beta$, and $\sum_{k=1}^{4}\rho_k = 1$, (13) can be rewritten as follows:

$$\dot{\mathbf{x}} = \left(\sum_{k=1}^{4}\rho_k\right)(\mathbf{A}_0\mathbf{x} + \mathbf{B}_0 u) + (\rho_2+\rho_4)(\mathbf{A}_\alpha\mathbf{x} + \mathbf{B}_\alpha u) + (\rho_3+\rho_4)\mathbf{A}_\beta\mathbf{x}. \tag{19}$$

By regrouping (19) in the form of (14), one can prove that the values of $\mathbf{A}_k$ and $\mathbf{B}_k$ satisfy the expressions in the lemma. □

Notice that the linear combination in (14) is convex that in addition to $\sum_{k=1}^{4}\rho_k = 1$, all values of $\rho_k(t)$ fall between 0 and 1 due to $0 \le \alpha$ and $\beta \le 1$. A controller of the form

$$u = -\sum_{j=1}^{4}\rho_j\mathbf{K}_j\mathbf{x} \tag{20}$$

where the values of $\mathbf{K}_j$ are feedback gain matrices, which was proposed in [14] to stabilize the LPV system in (14). Yeh et al. [13] also applied the same control structure to stabilize the lateral dynamics of a bicycle under speed and acceleration variations. The controller therein is time-varying due to the presence of $\rho_{.}(t)$. Moreover, its implementation requires online computation of the convex combination of four linear, full-state-feedback controllers. In this work, considering that balancing control only serves as the lower level control for executing the commands from the upper level control, to reduce the overall control complexity, a simple constant-gain full-state-feedback law is adopted. The following theorem provides a synthesis method for computing the control gain matrix.

*Theorem 1 (Stabilization):* Given the polytopic system in (14) with the control law $u = -\mathbf{K}\mathbf{x}$, for $\mathbf{W} = \mathbf{W}^{\mathrm{T}} > \mathbf{0} \in \mathbb{R}^{3\times3}$ and $\lambda > 0 \in \mathbb{R}$, if there exists $\mathbf{Y} \in \mathbb{R}^{1\times3}$ satisfying the following LMIs:

$$(\mathbf{A}_k\mathbf{W} - \mathbf{B}_k\mathbf{Y}) + \left(\mathbf{W}\mathbf{A}_k^{\mathrm{T}} - \mathbf{Y}^{\mathrm{T}}\mathbf{B}_k^{\mathrm{T}}\right) + 2\lambda\mathbf{W} \le \mathbf{0} \tag{21}$$

for $k = 1$–$4$, then the closed-loop system is globally exponentially stable by setting

$$\mathbf{K} = \mathbf{Y}\mathbf{W}^{-1}. \tag{22}$$

Furthermore, $\|\mathbf{x}(t)\|_2$ is bounded by $C\|\mathbf{x}(0)\|_2 e^{-\lambda t}$ for some finite constant $C$.

*Proof:* For the Lyapunov function $V(\mathbf{x}) = \mathbf{x}^{\mathrm{T}}\mathbf{P}\mathbf{x}$ where $\mathbf{P} = \mathbf{W}^{-1}$, its time derivative along the closed-loop trajectory is

$$\dot{V} = \sum_{k=1}^{4}\rho_k\mathbf{x}^{\mathrm{T}}\left[\mathbf{P}(\mathbf{A}_k - \mathbf{B}_k\mathbf{K}) + \left(\mathbf{A}_k^{\mathrm{T}} - \mathbf{K}^{\mathrm{T}}\mathbf{B}_k^{\mathrm{T}}\right)\mathbf{P}\right]\mathbf{x}. \tag{23}$$

Substituting the gain matrices of (22) into $\dot{V}$ yields

$$\dot{V}(\mathbf{x}) = \sum_{k=1}^{4}\rho_k\mathbf{x}^{\mathrm{T}}\mathbf{P}\left[(\mathbf{A}_k\mathbf{W} - \mathbf{B}_k\mathbf{Y}) + \left(\mathbf{W}\mathbf{A}_k^{\mathrm{T}} - \mathbf{Y}^{\mathrm{T}}\mathbf{B}_k^{\mathrm{T}}\right)\right]\mathbf{P}\mathbf{x}. \tag{24}$$

By applying (21) and using the relations that $0 \le \rho_{(.)} \le 1$ and $\mathbf{P}\mathbf{W} = \mathbf{I}$, it can be derived that $\dot{V}$ is upper bounded by

$$\dot{V} \le -2\lambda\sum_{k=1}^{4}\rho_k\left[\mathbf{x}^{\mathrm{T}}\mathbf{P}\mathbf{x}\right] = -2\lambda\mathbf{x}^{\mathrm{T}}\mathbf{P}\mathbf{x} \tag{25}$$

in which the last equality is due to $\sum\rho_k = 1$. Therefore, $\dot{V} \le -2\lambda V$. The exponential stability of the closed-loop system is proved [15], and $(1/\lambda)$ can be viewed as the time constant for the closed-loop LPV system. □

Notice that the control law in the theorem is for stabilizing the bicycle's lateral dynamics around the equilibrium point $\mathbf{x} = \mathbf{0}$, so the bicycle tends to go straight. For the bicycle to perform cornering, the control system has to regulate both the velocity and the yaw rate while maintaining lateral stability. In this study, cornering control is posed as a regulation problem in which the bicycle state $\mathbf{x}$ is regulated to a reference state. The reference state, denoted by $\mathbf{x}_d = [\theta_d, \dot{\theta}_d, \delta_d]^{\mathrm{T}}$, is an equilibrium point of (11) when the bicycle moves at a constant command velocity $v_d$ and corners at a constant command yaw rate $\omega_d$, so it satisfies

$$\dot{\mathbf{x}}_d = \mathbf{A}_d\mathbf{x}_d + \mathbf{B}_d u_d = 0 \tag{26}$$

where $\mathbf{A}_d$ and $\mathbf{B}_d$ are defined by setting $v \equiv v_d$ in the system matrices $\mathbf{A}$ and $\mathbf{B}$ of (11), and $u_d$ is the corresponding reference input.

Examining the equality for the first component of $\dot{\mathbf{x}}_d$ in (26) gives $\dot{\theta}_d = 0$. By first substituting (8) into (3), and then replacing $\dot{\phi}$ and $v$, respectively, by $\omega_d$ and $v_d$ in the resulted equation, it yields the steering angle reference as follows:

$$\delta_d = \frac{1}{\sin\varepsilon_0}\tan^{-1}\left(\frac{\ell_b\omega_d}{v_d}\right). \tag{27}$$

The equalities for the third and second components of $\dot{\mathbf{x}}_d$ in (26) allow one to express $u_d$ and $\theta_d$ in terms of the computed $\delta_d$ by $u_d = (v_d/\ell_a)\delta_d$ and $\theta_d = -(v_d^2\sin\varepsilon_0/\ell_b g\delta_d)$. Hence, for cornering, the system should be regulated around the operating point

$$\mathbf{x}_d = \left[-\frac{v_d^2\sin\varepsilon_0}{\ell_b g}\delta_d \;\; 0 \;\; \delta_d\right]^{\mathrm{T}}. \tag{28}$$

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The following theorem proves that a similar design procedure as the stabilization problem can be applied to design the controller for state regulation.

*Theorem 2 (Regulation):* Given the bicycle dynamics in the polytopic form of (14) and the reference state vector defined in (28), the control law

$$u = -\mathbf{K}(\mathbf{x} - \mathbf{x}_d) + u_d \tag{29}$$

where $\mathbf{K}$ is computed by the stabilization theorem and $u_d = (v_d/\ell_a)\delta_d$ can make $\tilde{\mathbf{x}} = \mathbf{x}\text{-}\mathbf{x}_d$ that converges to zero exponentially.

*Proof:* First, the error dynamics can be derived as follows:

$$\begin{aligned}\dot{\tilde{\mathbf{x}}} &= \mathbf{A}\mathbf{x} + \mathbf{B}u - \mathbf{A}_d\mathbf{x}_d - \mathbf{B}_d u_d \\ &= \mathbf{A}\tilde{\mathbf{x}} + \mathbf{B}(u - u_d) + (\mathbf{A} - \mathbf{A}_d)\mathbf{x}_d + (\mathbf{B} - \mathbf{B}_d)u_d.\end{aligned}$$

By direct substitution, $(\mathbf{A} - \mathbf{A}_d)\mathbf{x}_d + (\mathbf{B} - \mathbf{B}_d)\mathbf{u}_d$ in the error dynamics can be reduced to $(v - v_d)\mathbf{b}$, where $\mathbf{b} = \begin{bmatrix} 0 & (v_d \sin\varepsilon_0/\ell_b h)\delta_d & (-(1/\ell_a) + c)\delta_d \end{bmatrix}^{\mathrm{T}}$. This, together with the control law in (29), yields the closed-loop error dynamics

$$\dot{\tilde{\mathbf{x}}} = (\mathbf{A} - \mathbf{B}\mathbf{K})\tilde{\mathbf{x}} + (v - v_d)\mathbf{b}. \tag{30}$$

Following similar derivation as the proof of the stabilization theorem but considering the existence of the residual term $(v - v_d)\mathbf{b}$, the time derivative of $V(\tilde{\mathbf{x}}) = \tilde{\mathbf{x}}^{\mathrm{T}} P \tilde{\mathbf{x}}$ can be obtained as follows:

$$\begin{aligned}\dot{V} &\le -2\lambda\tilde{\mathbf{x}}^{\mathrm{T}}\mathbf{P}\tilde{\mathbf{x}} + (v - v_d)\tilde{\mathbf{x}}^{\mathrm{T}}\mathbf{P}\mathbf{b} \\ &\le -2\lambda V + \frac{(v - v_d)\sqrt{V}}{\sqrt{\lambda_{\min}(\mathbf{P})}}\|\mathbf{P}\mathbf{b}\|_2\end{aligned} \tag{31}$$

In which the second equality is due to $\lambda_{\min}(\mathbf{P})\|\tilde{\mathbf{x}}\|_2^2 \le V$. Let $W = \sqrt{V}$, and use $\dot{W} = (\dot{V}/2\sqrt{V})$. For $V \neq 0$, (31) can be rewritten as follows:

$$\dot{W} \le -\lambda W + \frac{\|\mathbf{P}\mathbf{b}\|_2}{2\sqrt{\lambda_{\min}(\mathbf{P})}}(v - v_d). \tag{32}$$

Since, from Footnote 2, there exists a positive constant $C'$, so that $|v - v_d| \le C' e^{-c v_d t}$, by the comparison lemma in [16], if $c v_d \neq \lambda$, the evolution of $W(t)$ is upper bounded by a linear combination of exponentially converging functions as follows:

$$\begin{aligned}W(t) &\le W(0)e^{-\lambda t} + C''\int_0^{\mathrm{T}} e^{-\lambda\tau}e^{-c v_d(t-\tau)}d\tau \\ &\le W(0)e^{-\lambda t} + \frac{C''}{c v_d - \lambda}\left(e^{-\lambda t} - e^{-c v_d t}\right)\end{aligned}$$

where $C'' = (C'\|\mathbf{P}\mathbf{b}\|_2/2(\lambda_{\min}(\mathbf{P}))^{1/2})$. The exponential convergence of $\tilde{\mathbf{x}}$ is due to that $(\lambda_{\min}(\mathbf{P}))^{1/2}\|\tilde{\mathbf{x}}\|_2 \le W$. □

*Remark 1:* The formulation assumes $v_d$ and $\omega_d$ to be constants. Nevertheless, throughout the regulation process, the control system still exhibits LPV characteristics due to that $v$ is time-varying when following the acceleration rule in (12).

*Remark 2:* Although the stability proof is for the linearized model, since the Lyapunov function used is time-invariant, it can be argued that the equilibrium point of the original nonlinear model is locally asymptotically stable [15].

## IV. Robust Trajectory Control

The trajectory control is based on the unicycle model [17] describing the bicycle's kinematic behavior on the $XY$ plane

$$\begin{aligned}\dot{x} &= v\cos\phi \\ \dot{y} &= v\sin\phi \\ \dot{\phi} &= \omega\end{aligned} \tag{33}$$

in which $v$ and $\phi$ follow the definitions in Fig. 2, $[x, y]^{\mathrm{T}}$ is the position of the bicycle (point $O_r$) with respect to the inertial frame, and $\omega$ is the yaw rate. The trajectory control problem is to make the bicycle track the position and yaw angle of a virtual bicycle. Assuming the virtual bicycle is driven with assigned speed $v_r$ and yaw rate $\omega_r$, its position $[x_r, y_r]^{\mathrm{T}}$ and yaw rate $\phi_r$, which serve as the reference trajectory to the trajectory controller, are obtained by integrating the kinematics

$$\begin{aligned}\dot{x}_r &= v_r\cos\phi_r \\ \dot{y}_r &= v_r\sin\phi_r \\ \dot{\phi}_r &= \omega_r.\end{aligned} \tag{34}$$

Viewing $v$ and $\omega$ as the control inputs, to design the control laws for $v$ and $\omega$ to stabilize the tracking errors $\tilde{x} = x - x_r$, $\tilde{y} = y - y_r$, and $\tilde{\phi} = \phi - \phi_r$, we follow the control approach in [17] by first transforming the errors $\tilde{x}$ and $\tilde{y}$ in terms of the bicycle coordinates or defining the error state vector $\mathbf{e}$ as follows:

$$\mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{\phi} \end{bmatrix}. \tag{35}$$

Using (33) and (34), the error dynamics can be derived

$$\begin{aligned}\dot{\mathbf{e}} = &\begin{bmatrix} 0 & \omega_r & 0 \\ -\omega_r & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\mathbf{e} + \begin{bmatrix} 0 \\ \sin e_3 \\ 0 \end{bmatrix} v_r \\ &+ \begin{bmatrix} 1 & e_2 \\ 0 & -e_1 \\ 0 & 1 \end{bmatrix}\mathbf{u}\end{aligned} \tag{36}$$

where $\mathbf{u}$ is the new control input defined by

$$\mathbf{u} = \begin{bmatrix} v - v_r\cos e_3 \\ \omega - \omega_r \end{bmatrix}. \tag{37}$$

If linearized around $\mathbf{e} = \mathbf{0}$, (36) is reduced to

$$\dot{\mathbf{e}} = \begin{bmatrix} 0 & \omega_r & 0 \\ -\omega_r & 0 & v_r \\ 0 & 0 & 0 \end{bmatrix}\mathbf{e} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}\mathbf{u}. \tag{38}$$

Considering that $v_r$ and $\omega_r$ are time-varying, the error dynamics are also in the LPV form as (11) except that the scalar control input therein is replaced by a vectorial one. The error dynamics, thus, can be linearly parameterized as in (13) if the normalized variables $\alpha = (v_r - v_{\min})/(v_{\max} - v_{\min})$ and $\beta = (\omega_r - \omega_{\min})/(\omega_{\max} - \omega_{\min})$, where $v_{\max}(\omega_{\max})$ and $v_{\min}(\omega_{\min})$ are, respectively, the upper and lower bounds of $v_r(\omega_r)$, are defined. Thus, for trajectory following, one can apply Theorem 1 to compute the feedback matrix $\mathbf{K}$, so that the control law $\mathbf{u} = -\mathbf{K}\mathbf{e}$ can robustly stabilize the system of (38) under the time-varying $v_r$ and $\omega_r$.

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It should be noted that the trajectory control law determines how the velocity $v$ and yaw rate $\omega$ should vary in order for the bicycle to follow a given trajectory. Since $v$ and $\omega$ cannot be directly controlled but are, respectively, regulated through command velocity $v_d$ and command yaw rate $\omega_d$, the control law is expressed in terms of $v_d$ and $\omega_d$ as follows:

$$\begin{bmatrix} v_d \\ \omega_d \end{bmatrix} = -\mathbf{K}\mathbf{e} + \begin{bmatrix} v_r \cos e_3 \\ \omega_r \end{bmatrix}. \tag{39}$$

As depicted in Fig. 1, the trajectory control law produces $v_d$ and $\omega_d$. Given $v_d$, $v$ is obtained by integrating the acceleration rule (12) and serves as the reference command to the bicycle's speed control system. Given $\omega_d$, $\omega$ is regulated by the control law (29) with $\delta_d$ defined in (27). Because of (10), the control action $(u)$ can be used to compute the steering rate $\dot{\delta}$. By approximating $\dot{\delta}$ using $((\delta_{k+1} - \delta_k)/T)$, where $(\cdot)_k$ and $(\cdot)_{k+1}$ are, respectively, the current and the one-step-ahead values of the associated signal and $T$ is the sampling time, the one-step-ahead angle command to the position servo of the steering motor is given by the discrete approximation of (10) as follows:

$$\delta_{k+1} = \left[1 - T(1/\ell_a - c)v_k - Tc(v_d)_k\right]\delta_k + Tu_k \tag{40}$$

where $T$ is set as 0.01 s[3] in simulations and experiments.

The above discussion indicates that regulations of $v$ and $\omega$ can be treated as actuator dynamics to trajectory control system. To avoid performance degradation, the control design should have the actuator dynamics converge faster than the tracking dynamics. Therefore, in addition to the proposed stability theorems, simulations should also be used to tune the controller parameters to diminish the influence of actuator dynamics on tracking performance.

## V. Sensor Fusion

As shown in Fig. 1, the feedback signals for the low-level balancing and high-level trajectory controllers are provided by sensor fusion algorithms. For balancing control, an observer proposed in [18] computes the roll angle $\theta$ and roll rate $\dot{\theta}$ from the measurements of the wheel encoder and a six-axis inertial measurement unit (IMU). The trajectory control relies on an extended Kalman filter (EKF) to fuse encoder and IMU signals with GPS signal to obtain the position and orientation of the bicycle for feedback.

The motion model for EKF is the discretized version of (33) plus the velocity–acceleration relationship

$$\begin{aligned} x_{k+1} &= x_k + v_k \Delta t \cos\phi_k \\ y_{k+1} &= y_k + v_k \Delta t \sin\phi_k \\ \phi_{k+1} &= \phi_k + \omega_k \Delta t \\ v_{k+1} &= v_k + a_k \Delta t \end{aligned}$$

in which $\Delta t$ is the sampling period, $(\cdot)_k$ represents the quantity of interest sampled at the $k$th instant, and $\omega$ and $a$ are, respectively, the yaw rate and the longitudinal acceleration measured by the IMU. The measurement model is given by $\mathbf{z}_k = \begin{bmatrix} x_k & y_k & v_k \end{bmatrix}$ where the measurement vector $\mathbf{z}$ consists of the Cartesian coordinates provided by GPS sensor and velocity measurement from the wheel encoder.

[3]The time constants of the steering position servo, the balancing control, and the trajectory control in the subsequent simulations and experiments are, respectively, 0.09 s, 0.21 s [$\lambda = 4.7$ (rad/s)], and 0.83 s [$\lambda = 1.2$ (rad/s)], so the sampling time chosen provides a reasonable discrete approximation.

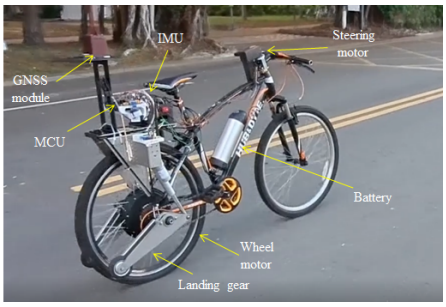



Fig. 4. Photograph of the prototype bicycle.

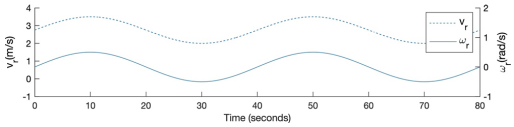



Fig. 5. Time-varying velocity and yaw rate of the virtual vehicle.

The EKF estimates the states by performing prediction using the motion model and update using the measurement model. The amount of update depends on the relative noise intensities between the model inputs ($a$ and $\omega$) and the measurement outputs $\mathbf{z}$. Due to different data rates in sensors, IMU prediction and the encoder update are run at 100 Hz and the GPS update at 5 Hz. Considering that the tracking control employed subsequently has a much slower time constant than sensors' data rates, the EKF leads to fast sensor dynamics, which has little impact on the closed-loop stability. Moreover, if the estimation errors due to EKF are treated as bounded disturbances to the control system, then the tracking errors are also bounded, since the tracking error dynamics for the unperturbed system are designed to be exponentially stable by Theorem 1 [19].

## VI. Self-Driving Control Validations

A prototype bicycle is constructed for experimental validation. The bicycle contains a 500-W brushless dc wheel motor as the rear wheel. The motor speed is regulated by a control unit, which contains a motor driver and an STM32F401RE MCU board. A Robotis MX-106R servo motor is installed on the pivot of the handlebar to provide steering action. An OpenIMU300ZA sensor containing a three-axis accelerometer and a three-axis gyro is attached to the bicycle frame. A Ublox C94-M8P RTK module, which performs real-time kinematic positioning, is installed on the back end of the bicycle to measure the GPS signal. The observer-based and the EKF sensor fusion, as well as the balancing and trajectory control algorithms, are implemented on another STM board. To provide support at stationary position and low speeds, the bicycle is also equipped with a set of landing gears, which can be actuated by linear electric actuators. The photograph of the prototype bicycle is shown in Fig. 4, and the relevant system parameters are as follows: $M = 23.1$ kg, $\ell_a = 0.3957$ m, $\ell_b = 1.053$ m, $h = 0.4338$ m, and $\varepsilon_0 = 70°$.

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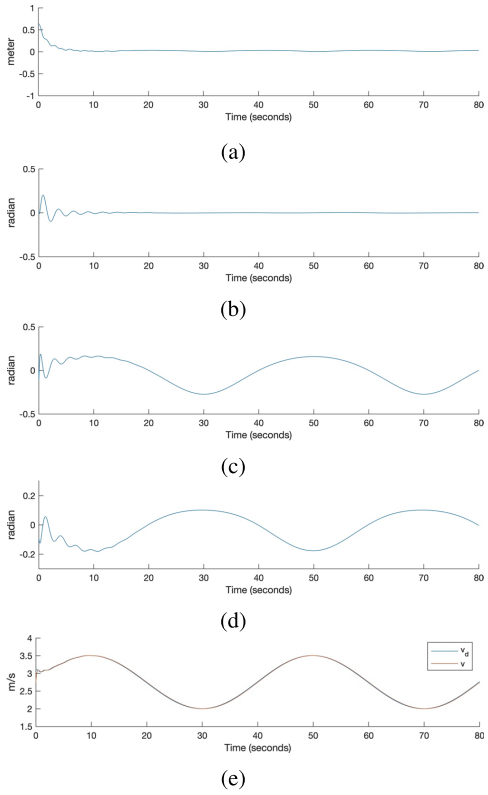



Fig. 6. Simulated responses for tracking and balancing. (a) Response of $(e_1^2 + e_2^2)^{1/2}$. (b) Response of $e_3$. (c) Response of $\delta$. (d) Response of $\theta$. (e) Responses of $v_d$ and $v$.

First, simulations are performed to examine the robustness of the proposed control system. The virtual bicycle is driven with $v_r(t) = 2.75 + 0.75\sin(t/20)$ (m/s) and $\omega_r(t) = 0.5\sin(t/20)$ (rad/s) whose trajectories are shown in Fig. 5. Given $v_r(t)$, $\omega_r(t)$, and $x_r(0) = y_r(0) = \phi_r(0) = 0$, the reference trajectory $[x_r(t), y_r(t), \phi_r(t)]^{\mathrm{T}}$ is computed by integrating (34). It is assumed that all the state measurements ($\theta$, $\dot{\theta}$, $\delta$, $x$, $y$, and $\phi$) are available, so the impact of sensor fusion on the control performance is not investigated in this case. The balancing control design assumes that $v_{\min}(v_{\max}) = 1.5$ (m/s)(4 (m/s)) and $c = 2.54$ (1/m). The feedback gain matrix computed by LMIs is $\mathbf{K} = [284.735, 58.262, -35.851]$. The trajectory control design assumes the same speed bounds as the balancing control, and the bounds for the yaw rate are set to $\pm 1$(rad/s). The feedback gain matrix computed is $\mathbf{K} = \begin{bmatrix} 1.574 & 0 & 0 \\ 0 & 4.603 & 4.907 \end{bmatrix}$.

Although both balancing and trajectory controllers are designed based on the linearized models, they are applied to the nonlinear models in (6) and (36) to simulate the control performance. In the simulation, the bicycle starts from $[x(0), y(0), \phi(0)]^{\mathrm{T}} = [-0.5\text{ m}, -0.4\text{ m}, 0\text{ rad}]^{\mathrm{T}}$. With both controllers in action, Fig. 6 shows the tracking errors, the steering angle, the roll angle, and the velocity responses. While the steering angle is modulated within 0.3 rad, the distance error $(e_1^2 + e_2^2)^{1/2}$ and the yaw angle error $e_3$ converge to zero in 5 s. Both the actual velocity $v$ and the command velocity $v_d$ exhibit sinusoidal characteristics as $v_r$, and there is no obvious difference between the two velocities, which provides justification on ignoring the actuator dynamics in stability analysis. The roll angle $\theta$ remains stable and exhibits sinusoidal behavior that reflects the characteristics of $\omega_r$. This simulation verifies that the proposed control system can provide robust regulation and robust tracking, respectively, under the time-varying $v$, $v_d$, and time-varying $v_r$, $\omega_r$.

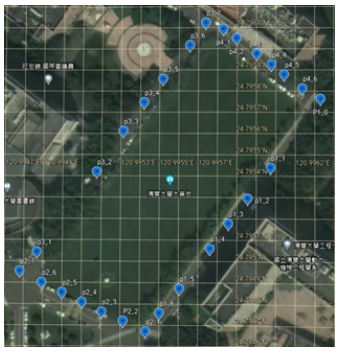


Fig. 7. Satellite image of the testing route.

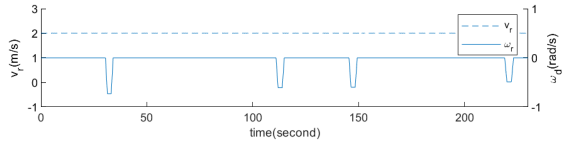


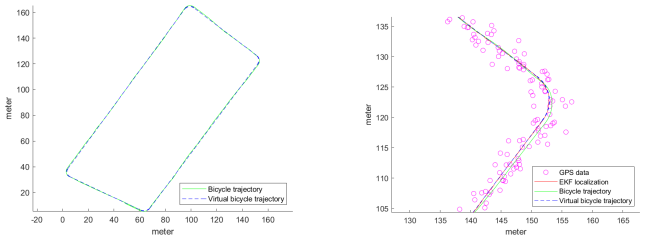

Fig. 8. Velocity and yaw rate of the virtual vehicle.

The next simulation is for the bicycle to drive on a testing route, which is a quadrilateral loop in the university campus, and its satellite image is shown in Fig. 7. For simplification, the route is decomposed into four straight and four arc segments. The virtual bicycle is assumed to be driven on the route with constant speed $v_r = 2$ m/s, and a trapezoidal trajectory shown in Fig. 8 is precomputed for the yaw-rate commanded $\omega_r$, so that the bicycle can complete the turns on the arc segments.

The control gain matrices in the previous simulation are adopted in this study. The feedback signals $x$, $y$, and $\phi$ for the trajectory controller are provided by the EKF whose design assumes that covariance matrices for the model inputs and measurement outputs are, respectively, diag[ $10^{-3}$ (rad$^2$/s$^2$) $10^{-3}$ (m$^2$/s$^4$) ] and diag[ 1.8 m$^2$ 1.8 m$^2$ $3.45 \times 10^{-3}$ (m$^2$/s$^2$) ]. The bicycle starts with an initial roll angle ($-5°$) and a position which is 0.834 m from the virtual bicycle. With comparable amounts of noises injected to the sensors, the simulation shows that throughout the control process, the distance error and the yaw angle error are kept within $8.22 \times 10^{-2}$ m and $4.14 \times 10^{-2}$ rad in rms. The paths of the virtual and the controlled bicycles are plotted in Fig. 9(a). The performance of EKF can be visualized from Fig. 9(b), where the close-up view of the top-right part of Fig. 9(a) with GPS measurements and the EKF localization results is shown. The EKF localization errors in rms are 0.27 m in $x$ and $y$, and $4.79 \times 10^{-2}$ rad in $\phi$.

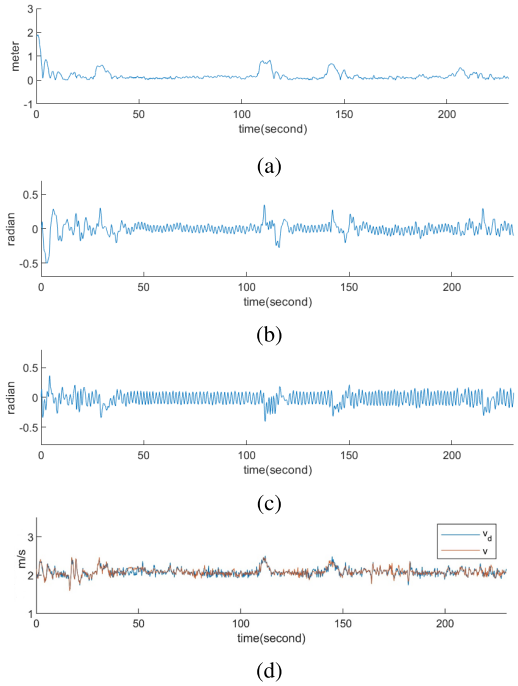
The control and sensor fusion algorithms in simulations are implemented, and the prototype bicycle is driven on the testing route. With the virtual bicycle moving as Fig. 8, the experimental responses are shown in Fig. 10. The control system is able to keep the rms distance and yaw angle errors to within 0.18 m and 0.07 rad for the entire path. Excluding the initial transience, the maximum distance and yaw angle errors are 0.64 m and 0.29 rad. Notice that the actuator dynamics due

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(a) (b)

Fig. 9. Simulated bicycle trajectories and GPS/EKF data. (a) Simulated paths of the virtual and the controlled bicycles. (b) Close-up view of (a) with GPS/EKF data.




(a) (b) (c) (d)

Fig. 10. Experimental responses for the proposed controller. (a) Response of $(e_1^2 + e_2^2)^{1/2}$. (b) Response of $e_3$. (c) Response of $\delta$. (d) Responses of $v_d$ and $v$.

to speed control servo have a time constant of about 0.2 s, so it converges about 4 times faster than the trajectory controller and should not cause tracking performance degradation. This is verified from Fig. 10(d), which shows that measured $v$ follows the reference $v_d$ closely. For comparison, we also implemented the pure pursuit method [20] with look-ahead distance equal to 10 m to perform path tracking at the designated speed.[4] The rms distance error is 0.62 m, which is about 3.4 times larger than the proposed tracking control. Video demonstration of the road test is given in https://youtu.be/zknZNQpsfgI.

## VII. Conclusion

This brief focuses on developing a control system for a self-driving bicycle, comprising low-level and high-level controllers for balancing and tracking, respectively. Given that both balancing and tracking dynamics display convex and time-varying characteristics, an LPV design is proposed for both controllers to ensure robustness. To reduce computational demands during simultaneous balancing and tracking control, the LPV method employs constant-gain full-state feedback for both controllers. Although constant gain control might appear conservative for LPV systems, the proposed theorem introduces a synthesis method to compute control gains that facilitates the design of tunable time constant in the closed-loop LPV system. Numerical and experimental verifications on a prototype bicycle demonstrate that autonomous driving on campus roads is successfully achieved. Future research is directed toward integrating MPC and obstacle detection to enhance collision avoidance and autonomous navigation capabilities of the self-driving bicycle.

[4]Using the symbols defined in Fig. 2, the pure pursuit method computes the steering angle by $\delta = (1/\sin\epsilon_0)\tan^{-1}((2\ell_b \sin\sigma)/\ell_d)$, in which $\ell_d$ is the look-ahead distance and $\sigma$ is the angle between $\overrightarrow{O_r O_f}$ and the vector connecting $O_r$ and the target point on the path whose distance from $O_r$ equals $\ell_d$.

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